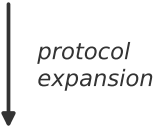


Protocol Hierarchy Expansion

Phase 1 of Stage 4 (4D Decomposition)

Original LRF Rules (first-match order)

r1: TCP 192.168.1.5 ANY 443 → ALLOW
r2: IP 192.168.0.0/16 ANY ANY → DENY



After Atomic Expansion (order preserved)

TCP: 192.168.1.5 / 443 → ALLOW (from r1)
TCP: 192.168.0.0/16 (rest) → DENY (from r2)
UDP: 192.168.0.0/16 → DENY (from r2)
ICMP: 192.168.0.0/16 → DENY (from r2; type [0, 255])

$\Pi(\text{IP}) = \{\text{TCP}, \text{UDP}, \text{ICMP}\}$ (super-set: r2 yields three sub-rules)

$\Pi(\text{TCP}) = \{\text{TCP}\}$ (already atomic: r1 yields one sub-rule)

Sub-rules inherit the position of their parent rule, so first-match priority is preserved within each atomic protocol class before the sweep-line phase.